Insurance Cost Analysis

```
insurance <- read.csv("data/medical_insurance.csv")
names(insurance)
```

```
 [1] "person_id"                    "age"
 [3] "sex"                          "region"
 [5] "urban_rural"                  "income"
 [7] "education"                    "marital_status"
 [9] "employment_status"            "household_size"
[11] "dependents"                   "bmi"
[13] "smoker"                       "alcohol_freq"
[15] "visits_last_year"             "hospitalizations_last_3yrs"
[17] "days_hospitalized_last_3yrs" "medication_count"
[19] "systolic_bp"                  "diastolic_bp"
[21] "ldl"                          "hba1c"
[23] "plan_type"                    "network_tier"
[25] "deductible"                   "copay"
[27] "policy_term_years"            "policy_changes_last_2yrs"
[29] "provider_quality"             "risk_score"
[31] "annual_medical_cost"          "annual_premium"
[33] "monthly_premium"              "claims_count"
[35] "avg_claim_amount"             "total_claims_paid"
[37] "chronic_count"                "hypertension"
[39] "diabetes"                     "asthma"
[41] "copd"                         "cardiovascular_disease"
[43] "cancer_history"               "kidney_disease"
[45] "liver_disease"                "arthritis"
[47] "mental_health"                "proc_imaging_count"
[49] "proc_surgery_count"           "proc_physio_count"
[51] "proc_consult_count"           "proc_lab_count"
[53] "is_high_risk"                 "had_major_procedure"
```

What proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors?

```r
#Ran a regression for modifiable life choices - might need to switch to glm if assumptions a
m1 <- lm(annual_premium ~ bmi + smoker + alcohol_freq, data=insurance)
summary(m1)
```

```

Call:
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)

Residuals:
    Min      1Q  Median      3Q     Max 
 -536.0  -220.4  -112.2    83.1 10204.1 

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)    
(Intercept)             655.3464     9.3629  69.994   <2e-16 ***
bmi                       3.4341     0.2497  13.754   <2e-16 ***
smokerFormer           -141.0053     4.6245 -30.491   <2e-16 ***
smokerNever            -192.4150     3.8799 -49.593   <2e-16 ***
alcohol_freqNone         -5.2296     6.0193  -0.869    0.385    
alcohol_freqOccasional   -5.8825     5.8747  -1.001    0.317    
alcohol_freqWeekly       -8.8204     6.2373  -1.414    0.157    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.3 on 99993 degrees of freedom
Multiple R-squared:  0.02619,	Adjusted R-squared:  0.02614 
F-statistic: 448.3 on 6 and 99993 DF,  p-value: < 2.2e-16
```

```r
# 1.DATA ANALYSIS
# View data structure
glimpse(insurance)
```

```
Rows: 100,000
Columns: 54
$ person_id                   <int> 75722, 80185, 19865, 76700, 92992, 76435, ~
$ age                         <int> 52, 79, 68, 15, 53, 63, 36, 21, 53, 28, 34~
$ sex                         <chr> "Female", "Female", "Male", "Male", "Male"~
$ region                      <chr> "North", "North", "North", "North", "Centr~
```

```
$ urban_rural                  <chr> "Suburban", "Urban", "Rural", "Suburban", ~
$ income                       <dbl> 22700, 12800, 40700, 15600, 89600, 305000,~
$ education                    <chr> "Doctorate", "No HS", "HS", "Some College"~
$ marital_status               <chr> "Married", "Married", "Married", "Married"~
$ employment_status            <chr> "Retired", "Employed", "Retired", "Self-em~
$ household_size               <int> 3, 3, 5, 5, 2, 3, 1, 3, 1, 4, 2, 2, 3, 4, ~
$ dependents                   <int> 1, 1, 3, 3, 0, 2, 0, 2, 0, 2, 1, 0, 1, 2, ~
$ bmi                          <dbl> 27.4, 26.6, 31.5, 31.6, 30.5, 20.4, 21.6, ~
$ smoker                       <chr> "Never", "Never", "Never", "Never", "Never~
$ alcohol_freq                 <chr> "None", "Weekly", "None", "None", "Daily",~
$ visits_last_year             <int> 2, 2, 1, 0, 3, 1, 0, 3, 0, 0, 8, 2, 2, 1, ~
$ hospitalizations_last_3yrs   <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ days_hospitalized_last_3yrs <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ medication_count             <int> 4, 3, 4, 1, 2, 2, 0, 1, 0, 0, 3, 3, 0, 2, ~
$ systolic_bp                  <dbl> 121, 131, 160, 104, 136, 137, 114, 82, 138~
$ diastolic_bp                 <dbl> 76, 79, 84, 68, 83, 96, 76, 64, 84, 62, 65~
$ ldl                          <dbl> 123.8, 97.3, 129.5, 160.3, 171.0, 69.2, 15~
$ hba1c                        <dbl> 5.28, 4.82, 5.51, 8.50, 5.20, 5.70, 5.54, ~
$ plan_type                    <chr> "PPO", "POS", "HMO", "HMO", "POS", "HMO", ~
$ network_tier                 <chr> "Bronze", "Gold", "Platinum", "Silver", "P~
$ deductible                   <int> 1000, 1000, 500, 500, 500, 500, 500, 2000,~
$ copay                        <int> 20, 10, 20, 20, 10, 20, 10, 20, 10, 10, 20~
$ policy_term_years            <int> 4, 1, 10, 5, 7, 5, 3, 6, 4, 4, 10, 9, 5, 2~
$ policy_changes_last_2yrs     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ provider_quality             <dbl> 3.73, 3.10, 3.90, 3.89, 3.90, 4.66, 4.30, ~
$ risk_score                   <dbl> 0.5714, 1.0000, 1.0000, 0.2857, 0.8681, 0.~
$ annual_medical_cost          <dbl> 6938.06, 1632.61, 7661.01, 5130.27, 1700.7~
$ annual_premium               <dbl> 876.05, 445.10, 1538.02, 820.63, 500.93, 5~
$ monthly_premium              <dbl> 73.00, 37.09, 128.17, 68.39, 41.74, 44.06,~
$ claims_count                 <int> 1, 4, 0, 0, 1, 1, 0, 2, 0, 0, 11, 2, 1, 3,~
$ avg_claim_amount             <dbl> 4672.59, 297.27, 0.00, 0.00, 1002.24, 1220~
$ total_claims_paid            <dbl> 4672.59, 1189.08, 0.00, 0.00, 1002.24, 122~
$ chronic_count                <int> 1, 2, 3, 1, 2, 1, 0, 1, 1, 0, 2, 1, 0, 0, ~
$ hypertension                 <int> 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, ~
$ diabetes                     <int> 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ asthma                       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ copd                         <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ cardiovascular_disease       <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, ~
$ cancer_history               <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ kidney_disease               <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ liver_disease                <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ arthritis                    <int> 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, ~
$ mental_health                <int> 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, ~
```

```
$ proc_imaging_count         <int> 1, 0, 1, 1, 2, 0, 1, 2, 0, 0, 0, 1, 0, 1, ~
$ proc_surgery_count         <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ proc_physio_count          <int> 2, 1, 2, 0, 1, 0, 1, 0, 1, 2, 1, 1, 1, 0, ~
$ proc_consult_count         <int> 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
$ proc_lab_count             <int> 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 2, 0, ~
$ is_high_risk               <int> 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, ~
$ had_major_procedure        <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
```

```r
summary(insurance)
```

```
   person_id           age              sex               region         
 Min.   :     1   Min.   :  0.00   Length:100000      Length:100000     
 1st Qu.: 25001   1st Qu.: 37.00   Class :character   Class :character  
 Median : 50000   Median : 48.00   Mode  :character   Mode  :character  
 Mean   : 50000   Mean   : 47.52                                        
 3rd Qu.: 75000   3rd Qu.: 58.00                                        
 Max.   :100000   Max.   :100.00                                        
 urban_rural            income          education         marital_status    
 Length:100000      Min.   :   1100   Length:100000      Length:100000     
 Class :character   1st Qu.:  21100   Class :character   Class :character  
 Mode  :character   Median :  36200   Mode  :character   Mode  :character  
                    Mean   :  49874                                        
                    3rd Qu.:  62200                                        
                    Max.   :1061800                                        
 employment_status  household_size    dependents          bmi       
 Length:100000      Min.   :1.000   Min.   :0.0000   Min.   :12.00  
 Class :character   1st Qu.:2.000   1st Qu.:0.0000   1st Qu.:23.60  
 Mode  :character   Median :2.000   Median :1.0000   Median :27.00  
                    Mean   :2.431   Mean   :0.8984   Mean   :26.99  
                    3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:30.40  
                    Max.   :9.000   Max.   :7.0000   Max.   :50.40  
    smoker          alcohol_freq       visits_last_year
 Length:100000      Length:100000      Min.   : 0.000  
 Class :character   Class :character   1st Qu.: 1.000  
 Mode  :character   Mode  :character   Median : 2.000  
                                       Mean   : 1.928  
                                       3rd Qu.: 3.000  
                                       Max.   :25.000  
 hospitalizations_last_3yrs days_hospitalized_last_3yrs medication_count
 Min.   :0.00000            Min.   : 0.0000             Min.   : 0.000  
 1st Qu.:0.00000            1st Qu.: 0.0000             1st Qu.: 0.000  
 Median :0.00000            Median : 0.0000             Median : 1.000  
```

```
 Mean   :0.09364           Mean   : 0.3734           Mean   : 1.236  
 3rd Qu.:0.00000           3rd Qu.: 0.0000           3rd Qu.: 2.000  
 Max.   :3.00000           Max.   :21.0000           Max.   :11.000  
  systolic_bp     diastolic_bp        ldl            hba1c       
 Min.   : 61.0   Min.   : 40.0   Min.   : 30.0   Min.   : 3.540  
 1st Qu.:107.0   1st Qu.: 67.0   1st Qu.: 99.4   1st Qu.: 5.160  
 Median :117.0   Median : 73.0   Median :120.0   Median : 5.440  
 Mean   :117.8   Mean   : 73.6   Mean   :120.0   Mean   : 5.606  
 3rd Qu.:128.0   3rd Qu.: 79.0   3rd Qu.:140.3   3rd Qu.: 5.760  
 Max.   :183.0   Max.   :114.0   Max.   :248.3   Max.   :11.940  
  plan_type         network_tier        deductible       copay      
 Length:100000      Length:100000      Min.   : 500   Min.   :10.00  
 Class :character   Class :character   1st Qu.: 500   1st Qu.:10.00  
 Mode  :character   Mode  :character   Median :1000   Median :20.00  
                                       Mean   :1227   Mean   :19.52  
                                       3rd Qu.:2000   3rd Qu.:30.00  
                                       Max.   :5000   Max.   :50.00  
 policy_term_years policy_changes_last_2yrs provider_quality   risk_score    
 Min.   : 1.000    Min.   :0.00000          Min.   :1.500    Min.   :0.0000  
 1st Qu.: 3.000    1st Qu.:0.00000          1st Qu.:3.200    1st Qu.:0.3297  
 Median : 6.000    Median :0.00000          Median :3.600    Median :0.5055  
 Mean   : 5.518    Mean   :0.05069          Mean   :3.599    Mean   :0.5198  
 3rd Qu.: 8.000    3rd Qu.:0.00000          3rd Qu.:4.010    3rd Qu.:0.7033  
 Max.   :10.000    Max.   :2.00000          Max.   :5.000    Max.   :1.0000  
 annual_medical_cost annual_premium    monthly_premium   claims_count   
 Min.   :   55.55    Min.   :  211.7   Min.   : 17.64   Min.   : 0.000  
 1st Qu.: 1175.12    1st Qu.:  352.1   1st Qu.: 29.34   1st Qu.: 0.000  
 Median : 2082.57    Median :  463.6   Median : 38.63   Median : 1.000  
 Mean   : 3009.45    Mean   :  582.3   Mean   : 48.53   Mean   : 1.622  
 3rd Qu.: 3707.96    3rd Qu.:  666.7   3rd Qu.: 55.56   3rd Qu.: 2.000  
 Max.   :65724.90    Max.   :10962.5   Max.   :913.55   Max.   :23.000  
 avg_claim_amount  total_claims_paid chronic_count     hypertension   
 Min.   :    0.0   Min.   :    0.0   Min.   :0.0000   Min.   :0.0000  
 1st Qu.:    0.0   1st Qu.:    0.0   1st Qu.:0.0000   1st Qu.:0.0000  
 Median :  318.0   Median :  642.5   Median :1.0000   Median :0.0000  
 Mean   :  656.5   Mean   : 1377.9   Mean   :0.7247   Mean   :0.2034  
 3rd Qu.:  872.2   3rd Qu.: 1795.5   3rd Qu.:1.0000   3rd Qu.:0.0000  
 Max.   :30010.5   Max.   :72517.9   Max.   :6.0000   Max.   :1.0000  
    diabetes           asthma             copd          cardiovascular_disease
 Min.   :0.00000   Min.   :0.00000   Min.   :0.00000   Min.   :0.00000       
 1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000       
 Median :0.00000   Median :0.00000   Median :0.00000   Median :0.00000       
 Mean   :0.08593   Mean   :0.05887   Mean   :0.03595   Mean   :0.05117       
```

```
 3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000
 Max.   :1.00000   Max.   :1.00000   Max.   :1.00000   Max.   :1.00000
 cancer_history    kidney_disease    liver_disease       arthritis
 Min.   :0.00000   Min.   :0.00000   Min.   :0.00000   Min.   :0.0000
 1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.00000   1st Qu.:0.0000
 Median :0.00000   Median :0.00000   Median :0.00000   Median :0.0000
 Mean   :0.02151   Mean   :0.01462   Mean   :0.01477   Mean   :0.1083
 3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.00000   3rd Qu.:0.0000
 Max.   :1.00000   Max.   :1.00000   Max.   :1.00000   Max.   :1.0000
 mental_health    proc_imaging_count proc_surgery_count proc_physio_count
 Min.   :0.0000   Min.   :0.0000     Min.   :0.0000     Min.   :0.0000
 1st Qu.:0.0000   1st Qu.:0.0000     1st Qu.:0.0000     1st Qu.:0.0000
 Median :0.0000   Median :0.0000     Median :0.0000     Median :0.0000
 Mean   :0.1301   Mean   :0.5085     Mean   :0.1587     Mean   :0.5084
 3rd Qu.:0.0000   3rd Qu.:1.0000     3rd Qu.:0.0000     3rd Qu.:1.0000
 Max.   :1.0000   Max.   :7.0000     Max.   :6.0000     Max.   :7.0000
 proc_consult_count proc_lab_count    is_high_risk    had_major_procedure
 Min.   :0.0000     Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.0000     1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.0000     Median :0.0000   Median :0.0000   Median :0.0000
 Mean   :0.5093     Mean   :0.5091   Mean   :0.3678   Mean   :0.1697
 3rd Qu.:1.0000     3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:0.0000
 Max.   :7.0000     Max.   :7.0000   Max.   :1.0000   Max.   :1.0000
```

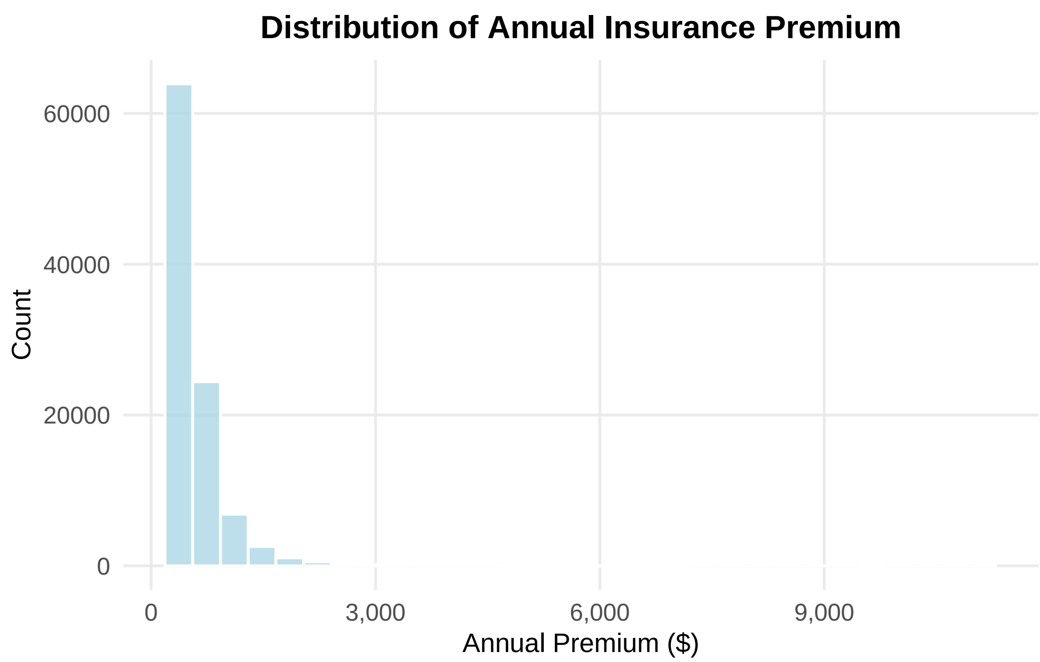
```r
# Check for missing values
colSums(is.na(insurance))
```

```
               person_id                      age
                       0                        0
                     sex                   region
                       0                        0
             urban_rural                   income
                       0                        0
               education           marital_status
                       0                        0
       employment_status           household_size
                       0                        0
              dependents                      bmi
                       0                        0
                  smoker             alcohol_freq
                       0                        0
        visits_last_year  hospitalizations_last_3yrs
```

```
                            0                         0
days_hospitalized_last_3yrs           medication_count
                            0                         0
                  systolic_bp              diastolic_bp
                            0                         0
                          ldl                     hba1c
                            0                         0
                    plan_type              network_tier
                            0                         0
                   deductible                     copay
                            0                         0
            policy_term_years  policy_changes_last_2yrs
                            0                         0
             provider_quality                risk_score
                            0                         0
           annual_medical_cost            annual_premium
                            0                         0
              monthly_premium              claims_count
                            0                         0
             avg_claim_amount         total_claims_paid
                            0                         0
                chronic_count              hypertension
                            0                         0
                     diabetes                    asthma
                            0                         0
                         copd    cardiovascular_disease
                            0                         0
               cancer_history            kidney_disease
                            0                         0
                liver_disease                 arthritis
                            0                         0
                mental_health        proc_imaging_count
                            0                         0
           proc_surgery_count         proc_physio_count
                            0                         0
           proc_consult_count            proc_lab_count
                            0                         0
                 is_high_risk      had_major_procedure
                            0                         0
```

```python
# univariate visualizations
```

```r
# Distribution of (annual_premium) using a histogram
ggplot(insurance, aes(x = annual_premium)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
  labs(title = "Distribution of Annual Insurance Premium",
       x = "Annual Premium ($)",
       y = "Count") +
  scale_x_continuous(labels = scales::comma) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```

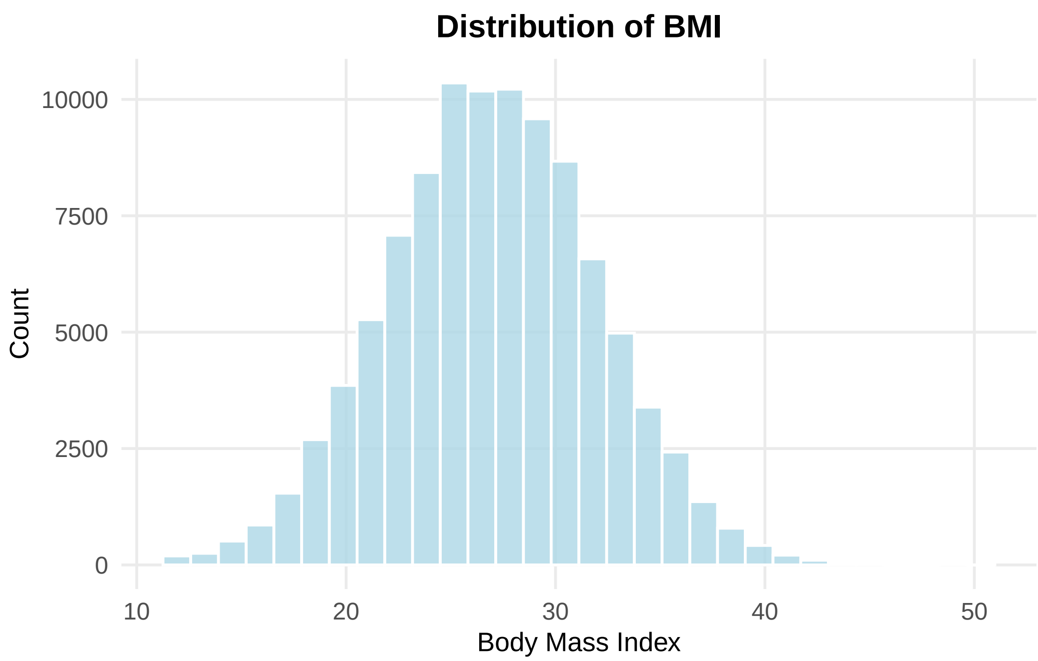



```r
# Summary statistics for premium
summary(insurance$annual_premium)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  211.7   352.1   463.6   582.3   666.7 10962.5
```

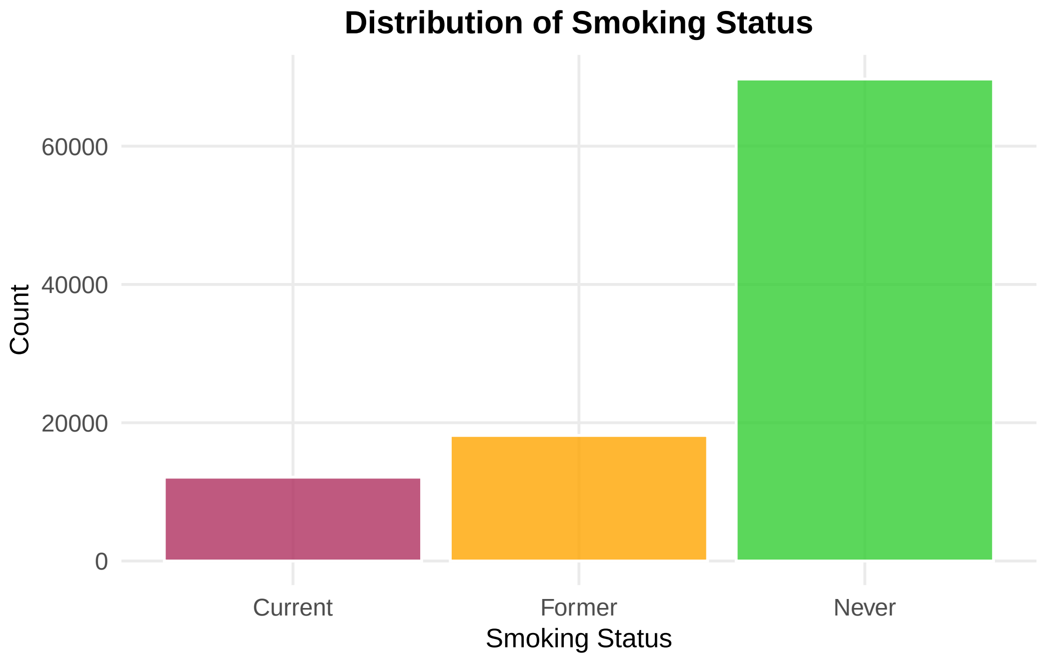
```r
# visualizations for distributions of modifiable lifestyle factors
# BMI visual
ggplot(insurance, aes(x = bmi)) +
```

```r
  geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
  labs(title = "Distribution of BMI",
       x = "Body Mass Index",
       y = "Count") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```

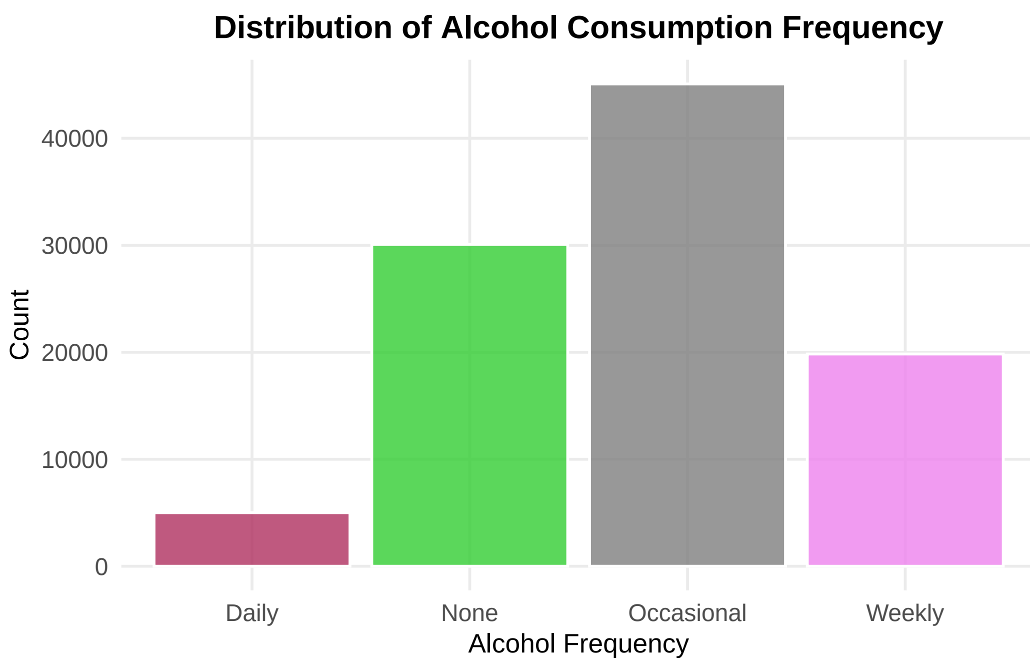



```r
# Smoking Status
ggplot(insurance, aes(x = smoker, fill = smoker)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Current" = "maroon",
                               "Former" = "orange",
                               "Never" = "limegreen")) +
  labs(title = "Distribution of Smoking Status",
       x = "Smoking Status",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
```

```
       panel.grid.minor = element_blank())
```




```
# Alcohol Frequency
ggplot(insurance, aes(x = alcohol_freq, fill = alcohol_freq)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Daily" = "maroon",
                               "None" = "limegreen",
                               "Occasionally" = "orange",
                               "Weekly" = "violet")) +
  labs(title = "Distribution of Alcohol Consumption Frequency",
       x = "Alcohol Frequency",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```




```r
ggplot(insurance, aes(x = smoker, y = annual_premium, fill = smoker)) +
  geom_violin(alpha = 0.6, color = "white") +
  geom_boxplot(width = 0.2, alpha = 0.8, color = "white", outlier.alpha = 0.5) +
  scale_fill_manual(values = c("Current" = "maroon",
                               "Former" = "orange",
                               "Never" = "limegreen")) +
  scale_y_continuous(
    labels = scales::comma,
    breaks = seq(0, 10000, 2000),
    expand = c(0, 0.02)
  ) +
  labs(title = "Insurance Premium by Smoking Status",
       subtitle = "Showing the primary effect of smoking on premiums",
       x = "Smoking Status",
       y = "Annual Premium ($)") +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
    plot.subtitle = element_text(size = 10, hjust = 0.5),
    axis.title = element_text(size = 10),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
```

```
)
```

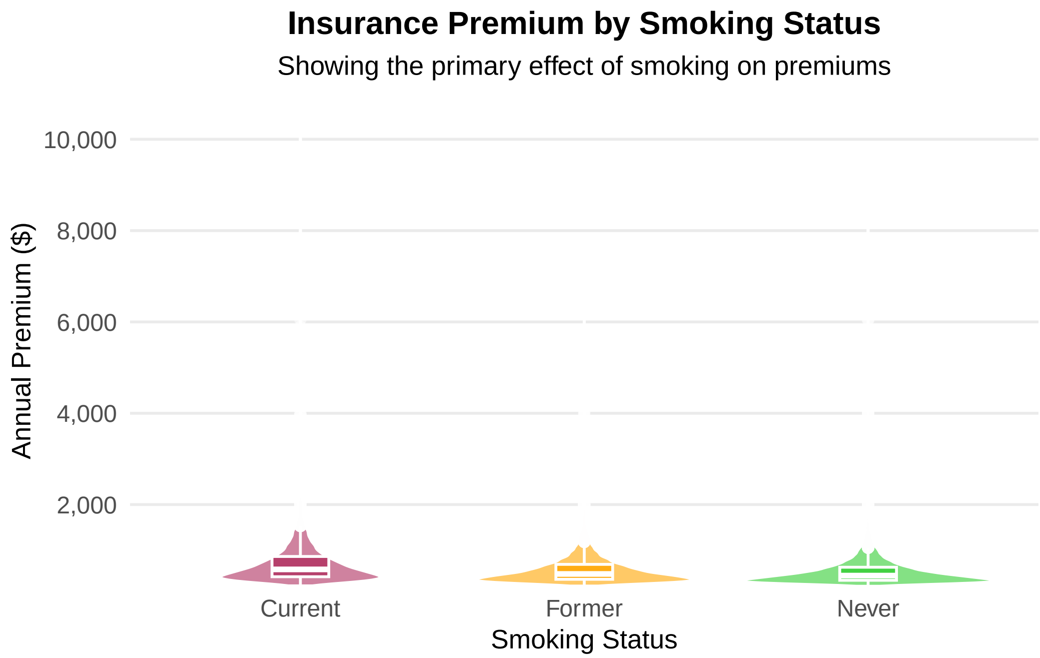


Figure [X] displays the distribution of annual insurance premiums across smoking status categories. The violin plots reveal substantial differences in premium distributions among the three groups. Current smokers demonstrate notably elevated premiums with a median around $1,200 and considerable variability extending to over $1,500 for the upper quartile. Former smokers show an intermediate distribution, with lower median premiums than current smokers but higher than never-smokers. Never-smokers exhibit the lowest and most concentrated premium distribution, with the majority of observations clustering below $1,000. The pronounced vertical separation between groups—particularly between current smokers and never-smokers—provides preliminary visual evidence that smoking status, a modifiable lifestyle factor, is strongly associated with insurance costs. The overlapping distributions suggest some shared variance with other factors, which will be explored through formal variance partitioning analysis.

```
# regression model stuff

#Modifiable lifestyle factors only
m1 <- lm(annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)

#Baseline demographic/socioeconomic factors only
m2 <- lm(annual_premium ~ age + sex + education + income +
         marital_status + household_size, data = insurance)
```

```r
#Full model (both modifiable and baseline factors)
m3 <- lm(annual_premium ~ bmi + smoker + alcohol_freq + age + sex +
         education + income + marital_status + household_size, data = insurance)

#Full model with interaction (BMI * smoker)
m4 <- lm(annual_premium ~ bmi * smoker + alcohol_freq + age + sex +
         education + income + marital_status + household_size, data = insurance)

#Display model summaries
cat("\n=== MODEL 1: Modifiable Factors Only ===\n")
```

```
=== MODEL 1: Modifiable Factors Only ===
```

```r
summary(m1)
```

```
Call:
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)

Residuals:
    Min      1Q  Median      3Q     Max 
 -536.0  -220.4  -112.2    83.1 10204.1 

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)    
(Intercept)             655.3464     9.3629  69.994   <2e-16 ***
bmi                       3.4341     0.2497  13.754   <2e-16 ***
smokerFormer           -141.0053     4.6245 -30.491   <2e-16 ***
smokerNever            -192.4150     3.8799 -49.593   <2e-16 ***
alcohol_freqNone         -5.2296     6.0193  -0.869    0.385    
alcohol_freqOccasional   -5.8825     5.8747  -1.001    0.317    
alcohol_freqWeekly       -8.8204     6.2373  -1.414    0.157    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.3 on 99993 degrees of freedom
Multiple R-squared:  0.02619,   Adjusted R-squared:  0.02614 
F-statistic: 448.3 on 6 and 99993 DF,  p-value: < 2.2e-16
```

```r
cat("\n=== MODEL 2: Baseline Factors Only ===\n")
```

```
=== MODEL 2: Baseline Factors Only ===
```

```r
summary(m2)
```

```
Call:
lm(formula = annual_premium ~ age + sex + education + income +
    marital_status + household_size, data = insurance)

Residuals:
    Min      1Q  Median      3Q     Max
 -488.2  -223.2  -114.1    83.2 10245.2

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            4.268e+02  7.089e+00  60.210   <2e-16 ***
age                    3.186e+00  7.839e-02  40.638   <2e-16 ***
sexMale                2.538e+00  2.532e+00   1.002   0.3162
sexOther              -1.051e+00  9.012e+00  -0.117   0.9072
educationDoctorate    -7.176e+00  7.663e+00  -0.936   0.3490
educationHS            3.658e+00  3.455e+00   1.059   0.2897
educationMasters       7.065e+00  4.104e+00   1.722   0.0851 .
educationNo HS         1.285e+01  6.024e+00   2.132   0.0330 *
educationSome College  3.972e+00  3.445e+00   1.153   0.2489
income                -5.740e-05  2.678e-05  -2.143   0.0321 *
marital_statusMarried -1.665e+00  5.213e+00  -0.319   0.7494
marital_statusSingle  -1.915e+00  5.185e+00  -0.369   0.7119
marital_statusWidowed -3.340e+00  7.829e+00  -0.427   0.6697
household_size         1.690e+00  1.318e+00   1.282   0.1998
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 396.3 on 99986 degrees of freedom
Multiple R-squared:  0.01641,   Adjusted R-squared:  0.01628
F-statistic: 128.3 on 13 and 99986 DF,  p-value: < 2.2e-16
```

```r
cat("\n=== MODEL 3: Full Model ===\n")
```

```
=== MODEL 3: Full Model ===
```

```r
summary(m3)
```

```
Call:
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq + age +
    sex + education + income + marital_status + household_size,
    data = insurance)

Residuals:
    Min      1Q  Median      3Q     Max
 -621.0  -214.3  -106.9    82.9 10068.3

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             4.996e+02  1.155e+01  43.239   <2e-16 ***
bmi                     3.412e+00  2.475e-01  13.784   <2e-16 ***
smokerFormer           -1.412e+02  4.585e+00 -30.785   <2e-16 ***
smokerNever            -1.936e+02  3.847e+00 -50.318   <2e-16 ***
alcohol_freqNone       -4.573e+00  5.968e+00  -0.766   0.4436
alcohol_freqOccasional -5.761e+00  5.825e+00  -0.989   0.3226
alcohol_freqWeekly     -9.452e+00  6.184e+00  -1.528   0.1264
age                     3.214e+00  7.734e-02  41.562   <2e-16 ***
sexMale                 3.333e+00  2.498e+00   1.334   0.1821
sexOther               -9.985e-01  8.891e+00  -0.112   0.9106
educationDoctorate     -7.615e+00  7.559e+00  -1.007   0.3138
educationHS             3.581e+00  3.408e+00   1.051   0.2934
educationMasters        7.980e+00  4.048e+00   1.971   0.0487 *
educationNo HS          1.322e+01  5.943e+00   2.224   0.0261 *
educationSome College   4.348e+00  3.398e+00   1.280   0.2007
income                 -5.671e-05  2.642e-05  -2.146   0.0318 *
marital_statusMarried  -1.934e+00  5.143e+00  -0.376   0.7069
marital_statusSingle   -1.619e+00  5.115e+00  -0.317   0.7516
marital_statusWidowed  -5.039e+00  7.723e+00  -0.652   0.5141
household_size          1.556e+00  1.301e+00   1.197   0.2315
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 391 on 99980 degrees of freedom
Multiple R-squared:  0.04291,	Adjusted R-squared:  0.04273 
F-statistic: 235.9 on 19 and 99980 DF,  p-value: < 2.2e-16
```

```r
cat("\n=== MODEL 4: Full Model + Interaction ===\n")
```

```
=== MODEL 4: Full Model + Interaction ===
```

```r
summary(m4)
```

```
Call:
lm(formula = annual_premium ~ bmi * smoker + alcohol_freq + age + 
    sex + education + income + marital_status + household_size, 
    data = insurance)

Residuals:
    Min      1Q  Median      3Q     Max 
 -622.6  -214.3  -106.9    82.9 10065.5 

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)    
(Intercept)             4.864e+02  2.123e+01  22.913  < 2e-16 ***
bmi                     3.904e+00  7.100e-01   5.499 3.82e-08 ***
smokerFormer           -1.214e+02  2.513e+01  -4.831 1.36e-06 ***
smokerNever            -1.797e+02  2.103e+01  -8.547  < 2e-16 ***
alcohol_freqNone       -4.596e+00  5.968e+00  -0.770   0.4413    
alcohol_freqOccasional -5.779e+00  5.825e+00  -0.992   0.3212    
alcohol_freqWeekly     -9.475e+00  6.184e+00  -1.532   0.1255    
age                     3.214e+00  7.734e-02  41.558  < 2e-16 ***
sexMale                 3.321e+00  2.498e+00   1.329   0.1838    
sexOther               -1.032e+00  8.891e+00  -0.116   0.9076    
educationDoctorate     -7.647e+00  7.559e+00  -1.012   0.3117    
educationHS             3.587e+00  3.408e+00   1.052   0.2927    
educationMasters        7.979e+00  4.048e+00   1.971   0.0487 *  
educationNo HS          1.324e+01  5.943e+00   2.227   0.0259 *  
educationSome College   4.343e+00  3.398e+00   1.278   0.2012    
income                 -5.674e-05  2.642e-05  -2.148   0.0317 *  
marital_statusMarried  -1.934e+00  5.143e+00  -0.376   0.7069    
```

```
marital_statusSingle   -1.609e+00  5.115e+00  -0.314   0.7531
marital_statusWidowed  -5.056e+00  7.723e+00  -0.655   0.5127
household_size          1.553e+00  1.301e+00   1.194   0.2325
bmi:smokerFormer       -7.345e-01  9.171e-01  -0.801   0.4232
bmi:smokerNever        -5.149e-01  7.694e-01  -0.669   0.5033
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391 on 99978 degrees of freedom
Multiple R-squared:  0.04292,   Adjusted R-squared:  0.04271
F-statistic: 213.5 on 21 and 99978 DF,  p-value: < 2.2e-16
```

We fit four linear regression models to examine how modifiable lifestyle factors and baseline demographic characteristics relate to insurance premiums. Model 1 included only modifiable factors (BMI, smoking status, alcohol frequency), Model 2 included only baseline factors (age, sex, education, income, marital status, household size), Model 3 combined both sets of predictors, and Model 4 added a BMI by smoking interaction term.

Model 1 explained 2.6% of premium variance and was statistically significant ($F = 448.3$, $p < 0.001$). Smoking status was the strongest predictor. Compared to current smokers, former smokers paid \$141 less annually and never smokers paid \$192 less (both $p < 0.001$), representing roughly 20 to 26% lower premiums. BMI showed a positive association, with each unit increase linked to a \$3.43 premium increase ($p < 0.001$). Alcohol consumption was not significantly related to premiums at any level (all $p > 0.15$).

Model 2 explained only 1.6% of premium variance. Age was the strongest baseline predictor, with each year associated with a \$3.19 premium increase ($p < 0.001$). Income showed a small negative effect ($p = 0.032$), while sex, marital status, and household size were not significant.

Model 3 combined all predictors and explained 4.3% of variance, nearly the sum of Models 1 and 2. Key predictors maintained stable effect sizes. Smoking coefficients were nearly identical to Model 1 (former: \$141 less, never: \$194 less, both $p < 0.001$), as were BMI and age effects. This stability suggests minimal confounding between modifiable and baseline factors.

Model 4 tested for BMI by smoking interactions but explained virtually the same variance as Model 3 (4.29% vs 4.27%). The interaction terms were not significant ($p = 0.42$ and $p = 0.50$), indicating the relationship between BMI and premiums does not differ across smoking groups. We therefore selected Model 3 as our final model.

The low R-squared values across all models (1.6% to 4.3%) indicate that most premium variance is unexplained by measured predictors. This likely reflects unmeasured factors such as coverage type, deductibles, policy terms, or company-specific pricing. However, the significant effects of modifiable factors, particularly smoking status, demonstrate that lifestyle choices contribute meaningfully to premium determination.